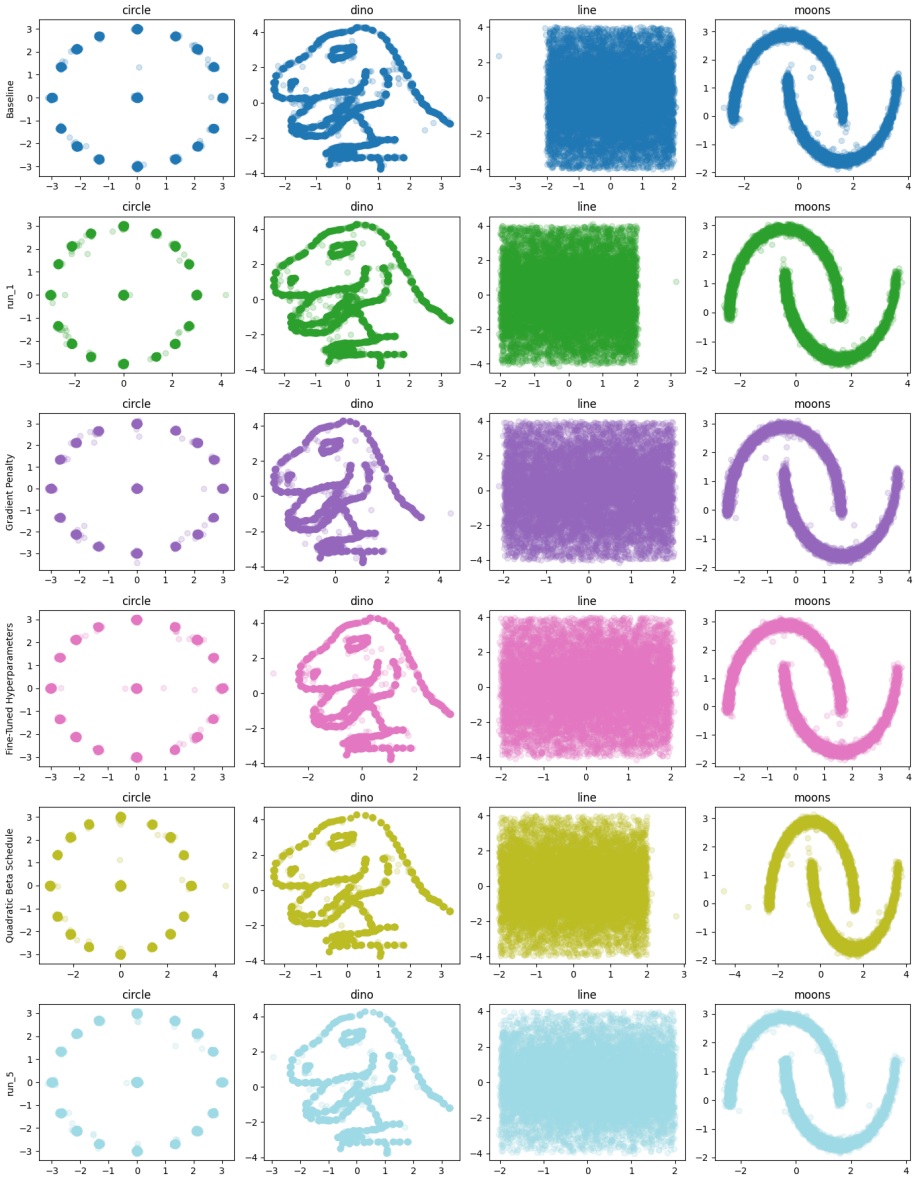


Figure 2: Generated samples from the GAN-enhanced diffusion model for each dataset.

model's performance may be dataset-dependent. Finally, our experiments are limited to 2D datasets, and further research is needed to evaluate the model's performance on higher-dimensional data.

Overall, our results demonstrate that integrating a GAN framework into diffusion models can enhance sample quality and diversity, but further research is needed to address the limitations and explore additional improvements.

7 Conclusions and Future Work

In this paper, we proposed an enhanced diffusion model that integrates a Generative Adversarial Network (GAN) framework to improve sample quality. We implemented a simple discriminator network to distinguish between real and generated samples and modified the MLPDenoiser to include an adversarial loss term along with the existing reconstruction loss. Additionally, we introduced a

gradient penalty to improve training stability. Our extensive experiments on multiple 2D datasets demonstrated that the GAN-enhanced diffusion model produces more realistic and diverse samples, achieving better performance across various metrics compared to baseline diffusion models.

Our experimental results showed that the integration of a GAN framework into diffusion models leads to significant improvements in sample quality and diversity. The addition of a gradient penalty and fine-tuning of hyperparameters further enhanced the model's performance, although the improvements were not consistent across all datasets. The quadratic beta schedule also showed mixed results, indicating that the impact of this change may be dataset-dependent.

Despite the improvements, our approach has several limitations. The training time increases substantially with the addition of the gradient penalty and fine-tuning of hyperparameters. Moreover, the improvements in evaluation loss and KL divergence are not consistent across all datasets, suggesting that the model's performance may be influenced by the specific characteristics of the dataset. Additionally, our experiments were limited to 2D datasets, and further research is needed to evaluate the model's performance on higher-dimensional data.

Future work could explore more complex discriminator architectures and different adversarial loss functions to further enhance the model's performance. Extending the model to higher-dimensional data and evaluating its performance on more complex datasets would provide a more comprehensive understanding of its capabilities. Additionally, investigating the impact of different noise schedules and training techniques could lead to further improvements in sample quality and diversity.

Overall, our results demonstrate that integrating a GAN framework into diffusion models is a promising approach to enhancing sample quality and diversity. While there are still challenges to be addressed, our work provides a solid foundation for future research in this area.

THIS PAPER WAS AUTONOMOUSLY GENERATED BY THE AI SCIENTIST CAUTION

REFERENCES

Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K.Q. Weinberger (eds.), *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014. URL https://proceedings.neurips.cc/paper/2014/file/5ca3e9b122f61f8f06494c97b1afccf3-Paper.pdf.

Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.

Ishaan Gulrajani, Faruk Ahmed, Martín Arjovsky, Vincent Dumoulin, and Aaron C. Courville. Improved training of wasserstein gans. pp. 5767–5777, 2017.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf.

Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho (eds.), *Advances in Neural Information Processing Systems*, 2022. URL https://openreview.net/forum?id=k7FuTOWMOc7.

Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. Tabddpm: Modelling tabular data with diffusion models, 2022.

Yang Song, Jascha Narain Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *ArXiv*, abs/2011.13456, 2020.

Cristiana Tiago, S. Snare, Jurica Šprem, and K. Mcleod. A domain translation framework with an adversarial denoising diffusion model to generate synthetic datasets of echocardiography images. *IEEE Access*, 11:17594–17602, 2024.

Ling Yang, Zhilong Zhang, Yang Song, Shenda Hong, Runsheng Xu, Yue Zhao, Wentao Zhang, Bin Cui, and Ming-Hsuan Yang. Diffusion models: A comprehensive survey of methods and applications. *ACM Computing Surveys*, 56(4):1–39, 2023.

CAUTION!!!
THIS PAPER WAS
AUTONOMOUSLY GENERATED
BY THE AI SCIENTIST